

Cancer-driving H3G34V/R/D mutations block H3K36 methylation and H3K36me3–MutS α interaction

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Somatic mutations on glycine 34 of histone H3 (H3G34) cause pediatric cancers, but the underlying oncogenic mechanism remains unknown. We demonstrate that substituting H3G34 with arginine, valine, or aspartate (H3G34R/V/D), which converts the non-side chain glycine to a large side chain-containing residue, blocks H3 lysine 36 (H3K36) dimethylation and trimethylation by histone methyltransferases, including SETD2, an H3K36-specific trimethyltransferase. Our structural analysis reveals that the H3 “G33–G34” motif is recognized by a narrow substrate channel, and that H3G34R/V/D mutations impair the catalytic activity of SETD2 due to steric clashes that impede optimal SETD2–H3K36 interaction. H3G34R/V/D mutations also block H3K36me3 from interacting with mismatch repair (MMR) protein MutS α , preventing the recruitment of the MMR machinery to chromatin. Cells harboring H3G34R/V/D mutations display a mutator phenotype similar to that observed in MMR-defective cells. Therefore, H3G34R/V/D mutations promote genome instability and tumorigenesis by inhibiting MMR activity.

histone mutation | SETD2 | histone methylation | mismatch repair

Histones are important protein components of chromatin. In addition to storing DNA and protecting it from environmental attacks, histones have emerged as critical factors regulating almost all DNA metabolic processes, including DNA replication, repair, and transcription (1). These important histone functions are executed by the highly sequence-conserved histone isoforms and their posttranslational modifications. For example, there are at least eight known H3 variants, including DNA replication-coupled H3.1 and transcription-essential H3.3 (1). Many H3 lysine residues can be posttranslationally modified and play critical roles in H3 functions. Trimethylation of H3K36 (H3K36me3) is well known for its role in active transcription (2, 3). Recent studies have revealed that H3K36me3 is essential for DNA repair (4–7), including DNA mismatch repair (MMR), a critical genome maintenance machinery that specifically corrects mispairs created during DNA replication (8–10). H3K36me3 interacts with the Pro-Trp-Trp-Pro (PWWP) domain of human mismatch recognition protein MutS α , an MSH2–MSH6 heterodimer. The interaction between H3K36me3 and the MutS α PWWP domain recruits MutS α to replicating chromatin (6), ensuring onsite mismatch removal. Depleting H3K36me3 or disrupting the H3K36me3–MutS α interaction leads to a mutator phenotype similar to that observed in cells with defects in MMR genes (6).

Consistent with the roles of H3K36me3 in genome maintenance, H3K36M and H3K36I mutations are associated with certain types of cancer, including chondroblastomas and pediatric glioblastomas (11–17). Somatic driver mutations for these cancers also include substitutions on glycine 34 of H3.3 (H3.3G34), particularly from glycine to arginine (H3.3G34R) and to valine (H3.3G34V) (11, 12, 15). We recently identified a histone H3.1 mutation in a colorectal cancer that converts glycine 34 to aspartate (H3.1G34D) (*SI Appendix, Fig. S1A and B*). The essential role of H3K36me3 in MMR indicates that H3K36 mutations cause

cancers; however, how H3G34 mutations induce tumorigenesis is unknown.

Since H3G34 is in close proximity to H3K36, we hypothesize that a large side chain created by G34D, G34R, and G34V mutations in H3 blocks the interaction between SETD2 and the H3 tail, inhibiting H3K36 trimethylation. Similarly, the large side chains may also inhibit the H3K36me3–MutS α interaction. We tested these hypotheses and found evidence of their validity. Cells carrying a heterozygous mutation at the G34 position display a partial mutator phenotype. Therefore, this study reveals that, like mutations on H3K36, mutations on H3G34 induce tumorigenesis by inhibiting MMR.

Results

H3G34D/V/R Mutations Block H3K36 Methylation in Vitro. To investigate whether H3G34 mutations affect H3K36 methylation, we performed an in vitro histone lysine methyltransferase (HKMT) assay using the purified human SETD2 catalytic domain (SETD2_{CD}; amino acid residues 1418–1714), synthesized H3 N-terminal tail peptides containing various K36 methylation statuses and a mutation on G34 (Fig. 1A), and tritium-labeled S-adenosylmethionine ([³H]-SAM), as described previously (18). As expected, SETD2_{CD} transferred a ³H-labeled methyl group efficiently to the wild-type (WT) H3 peptide containing a K36me2 (Fig. 1B, lane 3). Interestingly, ³H-labeled methyl groups were also added to peptides

Significance

Somatic mutations converting glycine 34 of histone H3 (H3G34) to a large side chain-containing residue (e.g., arginine, valine) cause pediatric gliomas; however, the mechanism of this is unknown. Because H3K36me3 is involved in mismatch repair (MMR) by recruiting MMR protein MutS α to chromatin, we hypothesized that H3G34R/V mutations block H3K36's interactions with both MutS α and H3K36-specific methyltransferases, leading to MMR deficiency. We show here that this is indeed the case. Structural analysis revealed that H3G34 resides in a narrow substrate channel of the H3K36 trimethyltransferase SETD2. Thus, H3G34R/V mutations impair the catalytic activity of SETD2. H3G34R/V mutations also block the H3K36me3–MutS α interaction. Cells harboring H3G34R/V mutations display a mutator phenotype. This study reveals the molecular basis of how H3G34 mutations cause pediatric gliomas.

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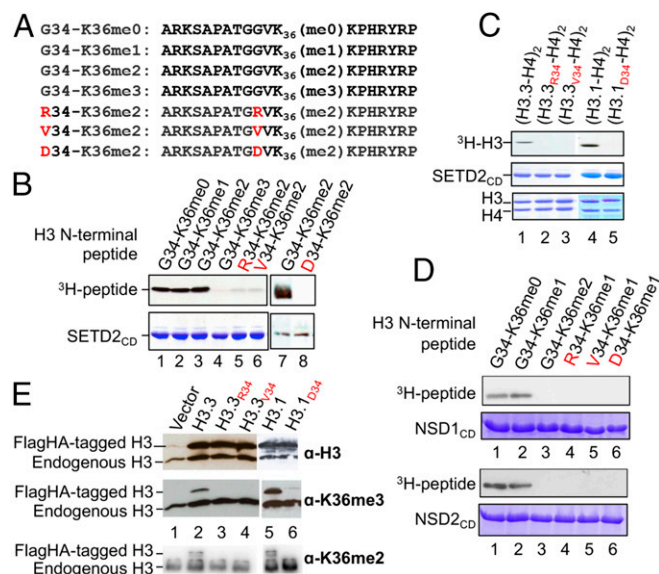


Fig. 1. H3G34 mutations block H3K36 methylation by SETD2. (A) Sequences of synthesized H3 peptides. Peptides with amino acids 25–43 of H3 were synthesized to contain one, two, or three methyl groups on K36 and WT G34 or R/V/D substitutions. (B and C) HKMT assay showing methylation capability of H3 peptides or H3-H4 tetramer by SETD2_{CD}. (D) HKMT assay demonstrating dimethylation capacity of H3 peptides by NSD1_{CD} or NSD2_{CD}. (E) Western blot analysis showing expression (Upper gel), H3K36 trimethylation (Middle gel), and H3K36 dimethylation (Lower gel) of native and ectopic H3 proteins with and without a mutated residue at the 34 position in HEK293 cells.

G34-K36me0 (Fig. 1B, lane 1) and G34-K36me1 (Fig. 1B, lane 2), indicating that SETD2_{CD} can also conduct monomethylation and dimethylation on K36 in vitro. This transfer reaction appears to be K36-specific, as no methylation on other lysine residues was detected in peptide G34-K36me3 (Fig. 1B, lane 4), in which K36 methylation is saturated. However, when G34 was replaced by arginine, valine, or aspartate, little ³H was incorporated into the peptide (Fig. 1B, lanes 5, 6, and 8). A similar assay was performed by substituting the H3 N-terminal peptide with (H3-H4)₂ tetramers. As in the peptide experiments, the WT (H3-H4)₂ tetramer could be methylated efficiently, but tetramers with G34R, G34V, and G34D mutations were not modified by SETD2_{CD} (Fig. 1C, lanes 2, 3, and 5).

We also examined the influence of G34 mutations on H3K36 dimethylation, which is the essential substrate for H3K36 trimethylation and can be catalyzed by the HKMTs NSD1 and NSD2. The results revealed that neither enzyme could transfer a methyl group to H3 N-terminal peptides containing R, V, or D at the 34 position (Fig. 1D, lanes 4–6), although both enzymes could methylate WT H3 peptide well (Fig. 1D, lanes 1–2). These results suggest that all H3G34V/R/D mutations block H3K36 methylation in vitro.

H3G34D/V/R Mutations Inhibit H3K36 Methylation in Vivo. To determine whether H3G34 mutations influence H3K36 methylation in vivo, we established HEK293 cell lines stably expressing flag-HA-tagged WT and mutant H3 proteins. The resulting cell lines were analyzed for H3K36 methylation of the ectopic H3 proteins. Both native and ectopic H3 proteins were expressed normally (Fig. 1E, Upper), and endogenous H3 in all cell lines could be recognized by an H3K36me3-specific antibody, indicating that H3K36 is trimethylated in these cells. However, when the same antibody was used to detect the methylation of ectopic H3, trimethylation of H3K36 was observed only in WT H3 proteins (Fig. 1E, Middle, lanes 2 and 5). We found similar results when analyzing H3K36 dimethylation (Fig. 1E, Lower). These results suggest that H3G34D/V/R mutations inhibit H3K36 methylation in vivo.

We next pulled down all ectopic H3 proteins using anti-flag beads and analyzed their H3K36 methylation levels by mass spectrometry. Methylation data were collected from both unsynchronized and G1-S phase synchronized cells. As shown in Table 1, H3K36me3 levels in both WT H3.1 and WT H3.3 proteins were higher in G1-S boundary cells than in unsynchronized cells, consistent with H3K36me3's role in recruiting MutSα in replicating cells (6, 19). Moreover, sufficient amounts (7.9–27%) of H3K36me3 were detected in WT H3 proteins (Table 1). However, substituting G34 with R, V, or D dramatically reduced H3K36me3 levels in both H3.1 and H3.3, with the highest level at 2.8% (Table 1). Reduced H3K36me2 levels were also observed in G34-mutated ectopic H3 proteins. These results, which are consistent with a recent study reporting inhibition of H3K36 methylation by H3G34L/W mutations (20), strongly support the notion that H3G34R/V/D mutations inhibit H3K36 methylation.

Structural Modeling Indicates Cis-Inhibition of SETD2 Activity by H3G34 Mutations. The crystal structure of the SETD2 catalytic domain in a complex with the H3 peptide shows that the G33-G34 of H3 is buried in the narrow substrate channel of SETD2 with restricted dimensions, which can fit only side chain-free glycine (Fig. 2A) (21). Structural modeling of SETD2 recognizing the H3 peptide and carrying G34 mutations (D/R/V) revealed that the mutated residues in position 34 impose a steric clash with the surrounding residues of SETD2, especially F1668, which mainly contributes to forming the inner wall of the substrate channel. In the case of G34D/R/V mutations, the channel can no longer fit the H3 substrate; that is, these H3G34 mutations block the H3 substrate binding of SETD2 (Fig. 2B). These structural analyses provide a molecular explanation of why H3G34D/R/V mutations inhibit H3K36 trimethylation in vivo and in vitro.

H3G34 Mutations Inhibit Interaction Between H3K36 and MSH6. To determine whether H3G34 mutations interfere with the H3K36me3–MutSα interaction, we conjugated biotinylated H3 N-terminal tail peptides with or without a G34 mutation (Fig. 1A) to streptavidin beads and used them to pull down recombinant (Fig. 3A) or native (Fig. 3B) MutSα. H3K36-trimethylated N-terminal tail peptides containing a G34 mutation, irrespective of G34D, G34R, or G34V, could precipitate only approximately 10–20% of the amount

Table 1. Effect of H3G34 mutations on H3K36 methylation in vivo

	Synchronized at G1-S, %					Unsynchronized, %				
	H3.3			H3.1		H3.3			H3.1	
H3K36 methylation status	G34R	G34V	WT	G34D	WT	G34R	G34V	WT	G34D	WT
None	45.8	93.8	57.6	82.9	50.0	56.2	52.3	37.3	91.8	30.8
Monomethylated	44.0	1.7	11.6	16.3	8.7	27.9	18.7	18.9	7.4	18.0
Dimethylated	9.4	3.7	16.2	0.8	14.2	14.7	26.2	34.5	0.8	43.3
Trimethylated	0.8	0.9	14.6	0.0	27.1	1.2	2.8	9.3	0.0	7.9

H3 proteins that were expressed ectopically with or without a G34 mutation were pulled down using anti-flag agarose beads. The resulting proteins were separated via SDS/PAGE and analyzed by mass spectrometry to determine the methylation level on K36.

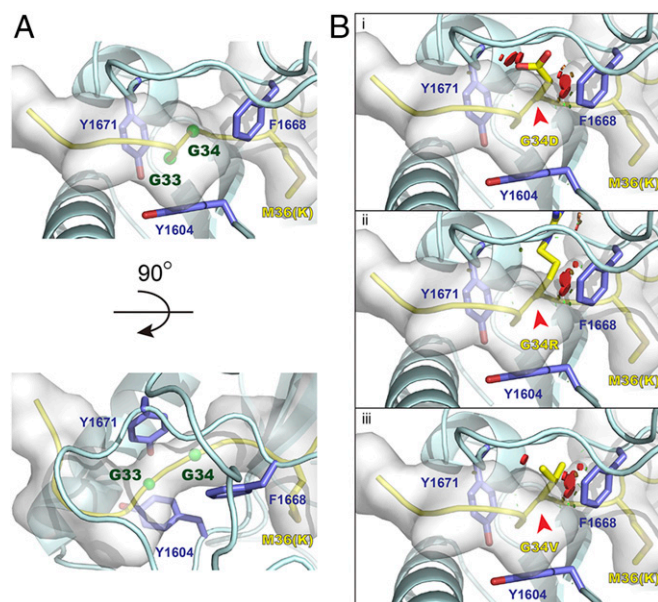


Fig. 2. Structural models showing accommodation of the H3G33-G34 fragment in the SETD2 substrate channel. SETD2 is shown in aquamarine; residues Y1604, F1668, and Y1671, composing the aromatic wall, are depicted as blue sticks. The H3 peptide is shown in a yellow and semitransparent-gray surface. (A) Crystal structure of SETD2 bound to the H3K36M mutant peptide (Protein Data Bank ID code 5JJY), showing the burial of H3G33-G34 in the SETD2 substrate channel. The side chain of M36(K) is shown as sticks, and the Cα of H3G33 and G34 are shown as green spheres. (B) Modeling of H3G34 mutants G34D (i), G34R (ii), and G34V (iii) placed within the SETD2 substrate channel. Mutant models are generated by PyMOL. Steric clashes are calculated using the PyMOL *show_bumps* script and shown as red plates; a higher number and diameter of the plates indicate heavier clash. G34D, G34R, and G34V mutants are shown as yellow sticks and indicated by red arrows.

of MutSα pulled down by K36-trimethylated WT peptides (Fig. 3*A* and *B*). Also, even though dimethylated K36 (K36me2) is not an optimal substrate for MutSα, the K36me2-G34-containing H3 peptide interacts with MutSα at least twice as actively as the peptide containing trimethylated K36 and G34V/R/D (Fig. 3*A* and *B*). These observations indicate that G34R/V/D mutations inhibit the H3K36me3–MutSα interaction *in vitro*.

To confirm that H3G34 mutations inhibit the H3K36me3–MutSα interaction *in vivo*, we measured the physical interactions between H3K36me3 and MutSα in cell lines with and without an H3G34 mutation using various approaches. First, a histone association assay (22) was conducted in two well-established pediatric glioblastoma cell lines, SF188 (H3.3-WT) and KNS42, with the latter harboring an H3.3G34V heterozygous mutation (*SI Appendix, Fig. S1C*) in *H3F3A* (12, 23). Although SF188 and KNS42 were derived from different patients, they have been widely used as a pair to study pediatric gliomas and histone mutations (12, 23, 24). The results show that the chromatin-bound MSH6 in KNS42 cells was only approximately 50% of that in SF188 cells (Fig. 3*C*). Similarly, chromatin-immunoprecipitation (ChIP) using an MSH6 antibody precipitated lower amounts of the H3 proteins in KNS42 cells than in SF188 cells (Fig. 3*D*). Consistent with these observations, confocal immunofluorescence analysis revealed that colocalization between H3K36me3 and MutSα was approximately 40% less in KNS42 cells than in SF188 cells (Fig. 3*E* and *F*), implying that less MutSα is recruited to chromatin in H3.3G34V KNS42 cells than in WT SF188 cells.

To verify that the reduced H3K36me3–MutSα interaction in KNS42 cells is due to the H3G34 mutation, we performed a histone association assay in SF188 and HEK293 cell lines stably expressing flag-tagged ectopic H3 proteins with or without an H3G34 mutation. As shown in Fig. 3*G* (SF188 cells) and Fig. 3*H*

(HEK293 cells), both ectopic H3 and endogenous H3 proteins were detected in the pellets, indicating that nucleosome compositions consist of both native and ectopic H3 proteins. These results suggest that a mutation at H3G34 does not prevent recruitment of H3 for nucleosome assembly. However, when an MSH6 antibody was used to detect the associated MutSα, reduced amounts of MSH6 were observed in all cell lines carrying an H3G34 mutation (Fig. 3*G* and *H*). Similar results were observed in a V5-tagged MSH6 pull-down experiment (Fig. 3*I*). These results suggest that H3.3G34 mutations cause less MutSα recruitment to chromatin, probably by inhibiting the H3K36me3–MutSα interaction.

We analyzed the published H3K36me3 ChIP-Seq data on SF188 and KNS42 cells (12) and found that approximately 35% of H3K36me3-enriched peaks in SF188 cells are not present in KNS42 cells (Fig. 4*A*). To determine whether the discrepancy in the H3K36me3 signal between these two cell lines is due to the different transcription profiles or to the difference in H3.3G34 mutations, we analyzed the transcriptome data of SF188 and KNS42 published previously (25). We found that only 25% (750/3,000) of the genes in the 35% group shown in Fig. 4*A* are significantly (fold change >2) down-regulated in KNS42 cells. In other words, the majority (75%) of the regions lacking the H3K36me3 signal in KNS42 cells could be related to the H3.3G34V mutation.

Since a lack of H3K36me3 enrichment can impact MutSα recruitment to chromatin (6), we performed ChIP-Seq analysis to determine the chromatin association of the MSH6 subunit of MutSα in these cells. We found that among the 35% of the KNS42 genomic regions missing H3K36me3 (Fig. 4*A*), 62% also lack the MSH6 signal (Fig. 4*B*). To show details of the affected regions in KNS42 cells, ChIP signals of H3K36me3 and MSH6 in representative “double-negative” regions in KNS42 cells were placed in a heatmap (with each horizontal position representing a genomic locus) and compared with those of the corresponding regions from SF188 cells (Fig. 4*C*). SF188 cells exhibited much higher levels of H3K36me3/MSH6 in these regions than KNS42 cells. To further verify this difference, we randomly selected three loci (*GSX2*, *KMT2A*, and *UBE4A*) and performed ChIP-qPCR analysis. The results showed significantly lower H3K36me3 and MSH6 signals in KNS42 cells than in SF188 cells (Fig. 4*D*). Taken together, our results suggest that H3G34 mutations interfere with the physical interaction between H3K36 and both SETD2 and MutSα.

Cells Harboring H3G34V/D Mutations Display a Weak Mutator Phenotype. Because H3G34 mutations inhibit both H3K36 methylation and the H3K36–MutSα interaction, we suspected that cells with somatic H3G34R/V/D mutations are partially defective in MMR. Thus, we examined microsatellite instability (MSI) in KNS42, SF188, and all five HEK293 cell lines stably expressing ectopic H3 proteins carrying H3.1G34D, H3.1WT, H3.3G34R, H3.3G34V, and H3.3WT mutations. As expected, no microsatellite pattern changes were detected in the clones derived from SF188 and HEK293 cells expressing ectopic WT H3.1 and H3.3 proteins. Surprisingly, we did not observe any new microsatellite species in any of the clones established from the KNS42 cells (with a heterozygous H3.3G34V mutation) or from cells with ectopic H3 proteins containing H3.3G34R/V mutations (*SI Appendix, Fig. S2*); however, we detected new microsatellite species in 2 out of 60 subclones derived from the H3.1G34D cell line (Fig. 5*A* and *B*). This discrepancy between H3.1G34D and H3.3G34R/V could be related to the fact that the G34D mutation imposes a more severe obstruction to H3K36me3 interactions with both MutSα and methyltransferases compared with other G34 mutations and/or because MMR primarily functions in replication, in which H3.1 is preferentially involved.

To determine whether cells with H3.3G34 mutations exhibit an elevated mutation frequency in random DNA sequences, we performed a hypoxanthine-guanine phosphoribosyltransferase (HPRT) mutability assay (26) in SF188 and KNS42 cells. This analysis

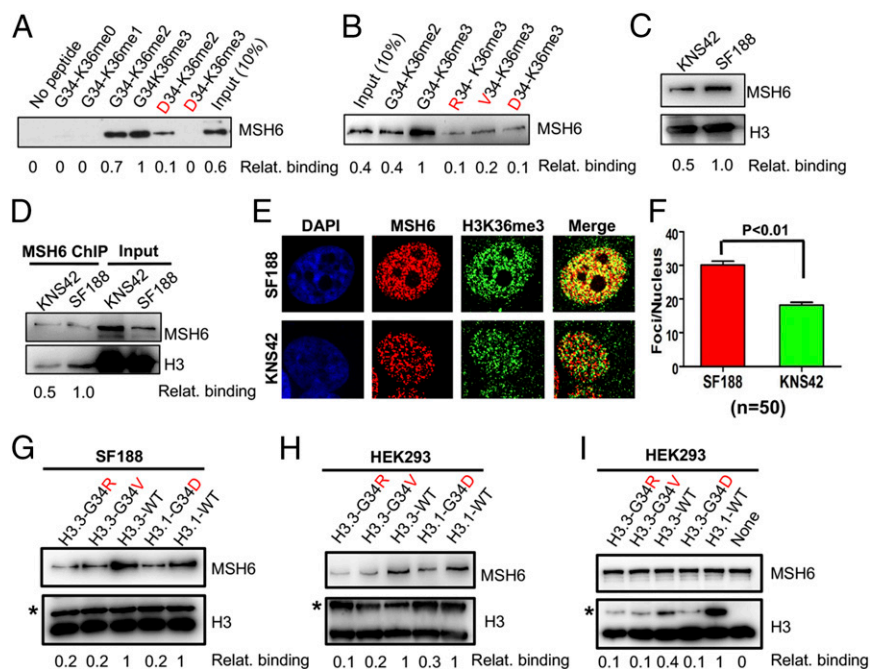


Fig. 3. H3G34 mutations inhibit the H3K36–MSH6 interaction. Recombinant (A) or nuclear extract (B) MutS α was pulled down by biotinylated H3 N-terminal tail peptides, and MutS α was visualized by Western blotting using an MSH6 antibody. (C) Chromatin fractionation assay combined with Western blotting to detect chromatin-bound MutS α in pediatric glioma cells. (D) ChIP assay using an MSH6 antibody, followed by Western blotting, to determine the H3–MutS α interaction in pediatric glioma cells. (E) Confocal immunofluorescence assays to determine MSH6 and H3K36me3 colocalization. (F) Quantification of MSH6 foci per nucleus in pediatric glioma cells. The “n” value indicates the number of nuclei analyzed, and the error bars represent the SEM. (G and H) Histone association assay to determine the H3–MutS α complex in SF188 (G) and HEK293 (H) cells expressing various ectopic H3 proteins. Flag-specific beads were used to pull down nucleosomes containing flag-tagged ectopic H3 (indicated by the asterisk), and MutS α was detected by an MSH6 antibody. (I) V5-tag pull-down assay to determine the H3–MutS α interaction as described in H. Band intensity was quantified using ImageJ software.

revealed a 19-fold higher mutation frequency in KNS42 cells compared with SF188 cells ($P < 0.01$) (Fig. 5C). Specific mutations in the “hot-spot” third exon of *HPRT* (26) in representative 6-thioguanine (6TG)-resistant clones from both lines are shown in *SI Appendix, Table S1*. We also conducted HPRT analysis in *MSH6*-knockdown SF188 cells (*SI Appendix, Fig. S1D*), which serve as a positive mutator control, and in all SF188 and HEK293 lines expressing WT H3 and H3G34-mutated H3. The results revealed that all cells expressing a H3G34-mutated H3 exhibited an approximately twofold-greater mutation frequency compared with their corresponding control cells, and *MSH6*-knockdown SF188 cells had a 7.7-fold greater mutation frequency compared with their control cells (Fig. 5C). These results suggest that H3G34 somatic mutations indeed cause an elevated mutation frequency.

It is noteworthy that the mutation frequency of KNS42 is approximately 10-fold higher compared with other H3G34-mutated cells. We reasoned that defects in other DNA repair pathways may contribute to the higher mutation frequency. Since SETD2-dependent H3K36me3 has also been shown to participate in homologous recombination (HR) repair of double-strand breaks (DSBs) by promoting DSB resection and allowing RAD51 binding to DNA damage sites (7), we measured the critical events/factors involved in hydroxyurea-induced HR repair of DSBs, including γ H2A.X foci formation and RAD51 recruitment (27, 28). As expected, DNA break formation (*SI Appendix, Fig. S3*), γ H2A.X foci formation (*SI Appendix, Fig. S4*), and recruitment of RAD51 to the damage sites (*SI Appendix, Fig. S4*) were all observed, but there was essentially no difference in the events/factors observed in KNS42 cells and in SF188 cells. Thus, the mutator phenotype observed in KNS42 cells is unrelated to HR repair (*Discussion*).

To further determine whether H3.3G34 mutations cause increased mutation frequencies, we analyzed whole-genome sequencing data of pediatric gliomas deposited in the International Cancer Genome Consortium (ICGC) database (15, 16). All tumor donors containing *H3F3A* mutations were divided into a G34 mutation group (mainly the H3.3G34R mutation in the dataset) and a non-G34 mutation group (including the H3.3K27M mutation). The total numbers of somatic mutations identified in each patient in these two groups are plotted in Fig. 5D. The results show that mutation densities were significantly higher ($P < 0.05$) in the H3G34 group than in the non-H3G34 group, although the latter group included two cases exhibiting a very high mutation

frequency, which may be related to uncharacterized mutations regulating an important genome maintenance system like MMR. Taken together, the data presented here strongly suggest that tumors with H3.3G34 mutations display a mutator phenotype.

Discussion

Somatic mutations of histone H3G34R/V/D are cancer-driving alterations for certain types of cancers, including pediatric gliomas (12, 15–17). However, the molecular mechanism by which these mutations promote tumorigenesis had not been defined until now. We have shown that these mutations execute their tumorigenic activity by inhibiting the MMR system, resulting in defects that cause cancer.

MMR in human cells relies on the H3K36me3 histone mark to recruit MutS α to chromatin. Depleting H3K36me3 or disrupting the H3K36me3–MutS α interaction leads to MMR defects and genome instability (6, 29). We found that cancer-driving H3G34D/R/V mutations obstruct MMR in at least two ways. First, these mutations prevent H3K36 dimethylation and trimethylation. Histone methyltransferase activities fail to methylate H3K36 in vivo and in vitro when H3 carries a D, R, or V substitution at the 34 position (Fig. 1 and Table 1). Cocystal structure analysis (21) revealed that the H3 “G33–G34” motif is recognized by a narrow substrate channel of the SETD2 catalytic domain, and that substituting G33–G34 with a residue containing a bulky sidechain (e.g., valine, arginine, aspartate) causes a steric clash with the channel and disrupts the interaction between the SETD2 cavity and H3K36 (Fig. 2), thereby blocking H3K36 trimethylation. We expect that H3K36 dimethylation by NSD1/2 is similarly blocked. Second, both pull-down and chromatin association assays revealed that H3G34 mutations reduce interactions between H3K36me3 and MutS α (Fig. 3A–D and G–I); fewer H3K36me3–MSH6 foci were also observed in glioma cells containing a somatic H3G34V mutation (Fig. 3E and F). Taken together, our data strongly support the notion that H3G34 mutations inhibit MMR by blocking H3K36 methylation and/or its interaction with MutS α .

Consistent with the fact that cells defective in MMR are hypermutable, an HPRT mutability assay and whole-genome sequencing analyses revealed that the H3G34-mutated cells and tumors display elevated mutation frequencies (Fig. 5C and D). We also detected MSI in cells expressing an H3.1 protein with a G34D mutation (Fig. 5A and B). However, unlike the H3.1G34D

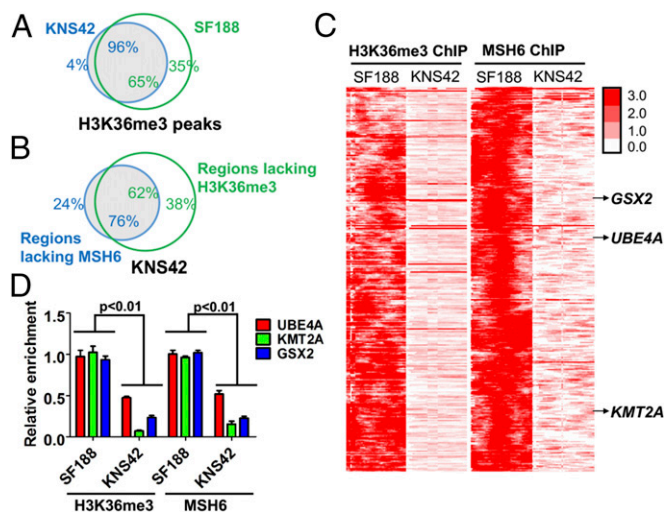


Fig. 4. H3G34V mutation alters chromatin distribution/enrichment of H3K36me3 and MSH6 in pediatric glioma cells. (A) Venn diagram illustrating H3K36me3 ChIP-Seq peaks in SF188 and KNS42 cell lines. The H3K36me3 ChIP-Seq data were from Bjerke et al. (12). (B) Venn diagram demonstrating overlapping regions that lack both H3K36me3 and MSH6 signals ("double negative") in KNS42 cells. The regions lacking H3K36me3 signals is the 35% portion shown in A. (C) Heatmap showing differential enrichment of H3K36me3 and MSH6 signals in the genetic loci corresponding to the double-negative regions between SF188 and KNS42 cells. Each horizontal line represents a specific genetic locus. The color intensity levels (from 0 to 3) indicate the ChIP signals. (D) Verification of H3K36me3 and MSH6 signals in three randomly selected loci (arrows in C) in SF188 and KNS42 cells using ChIP, followed by real-time PCR.

mutation-containing cells and typical MMR-deficient cells, H3.3G34 mutation-containing cells did not display MSI (*SI Appendix, Fig. S2*). This discrepancy may suggest that H3.1 and H3.3 differentially influence MMR.

Of note, KNS42 cells have an approximately 10-fold higher mutation frequency compared with the other H3G34-mutated cells that we investigated (Fig. 5C). It is possible that KNS42 cells are also defective in another genome maintenance pathway. We found that a defect in HR is not involved (*SI Appendix, Figs. S3 and S4*). However, the KNS42 cell line has been shown to be defective in O⁶-methylguanine methyltransferase (MGMT) (24). It is also known that loss of MGMT function renders cells resistant to 6-TG (30, 31), which serves as the mutability indicator of the HPRT assay. This might have contributed to the higher HPRT mutability observed in KNS42 cells. Another possibility may be related to the ratio of WT to G34-mutated H3 proteins in cells. Unlike SF188 and HEK293 cells, which express limited ectopic mutant H3 proteins, KNS42 cells express equal amounts of WT and mutant H3 proteins. Thus, compared with ectopically expressed H3 in SF188 and HEK293 cells, the mutated H3 in KNS42 cells has a greater likelihood of being recruited for nucleosome assembly to influence H3K36 methylation status and function. The lower mutation frequency in each of the G34-mutated HEK293 lines also may be related to the multiple X chromosomes in HEK293, which suggest its female origin (32). Although the HPRT assay relies on inactivating the only copy of X-linked *HPRT* in male-derived cells, the assay has been widely used to score mutation frequencies in female-derived cells as well, including HeLa (33), HEC-1-A (34), and HEC-59 (35) cells. However, the success of the latter application would require inactivating all X-linked *HPRT* copies at the same time, which may partially account for the low mutation frequencies observed in H3G34-mutated HEK293 cells.

In summary, we provide evidence suggesting that H3G34 mutations identified in cancers influence H3K36me3-mediated MMR for its role in stabilizing the genome. Under normal circumstances, SETD2 efficiently interacts with and trimethylates

H3.1K36 and H3.3K36. The resulting H3K36me3 recruits MutS α to chromatin. The chromatin-associated MutS α then performs its normal genome maintenance functions. However, when H3G34 is substituted with a large side chain residue, such as arginine, valine, or aspartate, it prevents H3K36 from fitting the cavity of the SETD2 catalytic domain (Fig. 2) and other histone methyltransferases, resulting in the inhibition of H3K36 methylation. Similarly, a large side chain at H3G34 also prevents H3K36 from interacting with MutS α , even if H3K36 is trimethylated. In either case, MutS α will not be recruited to chromatin, leading to MMR deficiency and eventually to tumorigenesis.

Materials and Methods

Cell Culture and Materials. Unless noted otherwise, all cells were cultured at 37 °C in a humidified atmosphere with 5% (vol/vol) CO₂. SF188 and KNS42 cells (kindly provided by Chris Jones, the Institute of Cancer Research, London) were cultured in DMEM (Invitrogen) supplemented with 10% FBS. Human embryonic kidney (HEK) 293 cells were used to establish lines stably expressing ectopic WT H3 or G34-mutated H3 proteins. MutS α was prepared as described previously (36). SETD2_{CD} (amino acid residues 1418–1714), NSD1_{CD} (residues 1852–2082), and NSD2_{CD} (residues 1011–1203) were cloned into pGEX-4T-2 vector (Novagen), expressed in *Escherichia coli*, and purified as GST-tagged proteins. Histone tetramers were constructed as described previously (37). H3 peptides were purchased from GenScript. All antibodies were obtained from commercial sources, including anti-H3 (Ab1791; Abcam), anti-H3K36me3 (Ab9050; Abcam), anti-MSH6 (BD Biosciences), anti- γ H2A.X (Cell Signaling Technology), anti-RAD51 (Cell Signaling Technology), and anti-flag (Bioeasy).

HKMT Assay. The HKMT assays were performed in 25- μ L reactions containing 1 μ g of SETD2_{CD}, NSD1_{CD}, or NSD2_{CD}; 1.5 μ M (0.15 μ Ci) [³H]-SAM cofactor; 5 μ g of peptide substrate or 2 μ g of reconstituted H3/H4 tetramer; 50 mM Hepes (pH 8.0); 0.005% Tween-20; 5 μ g/mL BSA; and 1 mM DTT. The reactions were incubated for 4 h at 30 °C, and the products were separated on an 18% SDS/PAGE gel. The ³H-labeled peptides were visualized by autoradiography.

Peptide Pull-Down Assay. The H3-MutS α pull-down assays were performed as described previously (6) in 200- μ L reactions containing 10 ng of biotinylated histone H3 peptides, 400 ng of MutS α or 100 μ g of HeLa nuclear extracts,

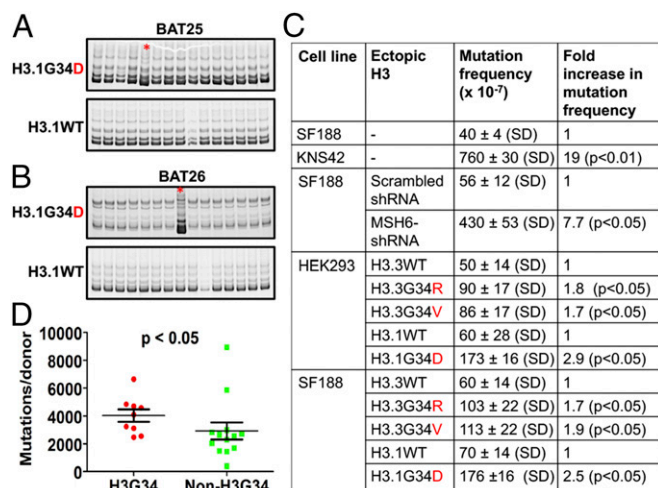


Fig. 5. Cells harboring H3G34 mutations display a mutator phenotype. (A and B) MSI assay of PCR products of microsatellite markers BAT25 (A) and BAT26 (B) in subclones derived from HEK293 cell lines expressing ectopic H3 proteins, as indicated. The red asterisks indicate the clones with new repeat patterns. (C) HPRT mutability assay in KNS42, SF188, and HEK293 cells expressing ectopic H3 with and without a G34 mutation, as indicated. The fold increase in mutation frequency was calculated using the mutation frequency of the corresponding control cells as a reference. SF188 cells with MSH6 knock-down by shRNA served as a positive control. (D) Comparison of the number of mutations in individual pediatric glioma patients with an H3.3G34 mutation (H3G34 group) and without an H3.3G34 mutation (non-H3G34 group). The data are from cancer genome studies deposited in the ICGC database (15, 16).

50 μ L of packed streptavidin bead slurry (Thermo Fisher Scientific), 50 mM Tris-HCl (pH 7.5), 300 mM NaCl, 0.1% Nonidet P-40, and 1 mM protease inhibitor. The mixtures were incubated at 4 °C for 1 h with rotation. After three washes with the reaction buffer, the bead-bound proteins were analyzed by SDS/PAGE, followed by Western blotting.

Chromatin Fractionation Preparation, Histone Association, and Immunofluorescence Analyses. Chromatin fractionation was performed as described previously (38), and the histone association assay was conducted as described previously (22). Immunofluorescence analysis was performed as described elsewhere (6). All experiments were performed using the same antibodies described above. Fluorescence images were obtained using a Zeiss Axio Observer Z1 inverted microscope and processed using ImageJ software.

MSI and HPRT Assays. MSI analysis was performed as described previously (39). For each cell line, 60 single colonies were established, and their genomic DNA was used to PCR-amplify five microsatellite markers: BAT25, BAT26, D2S123, D5S346, and D17S250. The PCR products were analyzed in denaturing PAGE. *HPRT* mutability assays were performed as described previously (6). Cells (1×10^5) were seeded in triplicate 100-mm Petri dishes for 12 h and fed with complete medium containing 15 μ M freshly prepared 6-TG. Plating efficiency was determined by seeding 1×10^3 cells without 6-TG. After 5 d of incubation, cell clones were cultured in 6-TG-free complete medium for another 10 d, followed by staining with 0.05% crystal violet. The mutation frequency was determined by dividing the number of 6-TG-resistant colonies by the total number of cells plated after correcting for their colony-forming ability.

ChIP-Seq Analysis. ChIP was performed as described previously (19, 40). Cells (2×10^7 SF188 and KNS42 cells) were harvested for ChIP experiments, and 4 μ g of antibodies against MSH6 (sc-10798; Santa Cruz Biotechnology) were used for immunoprecipitation. The ChIP products were subjected to sequencing library construction, and the resulting libraries were sequenced in single-end 50-bp (SE50) mode on the Illumina next-generation sequencing platform.

The sequencing reads underwent standard quality control pipeline validation, and clean reads were mapped to the human reference genome (UCSC hg19) using the SOAP2.21 alignment package (41) with default parameters. Reads mapped to more than one position were filtered out. Multiple reads mapping to the same position were counted only once to avoid PCR amplification bias. MACS software (42) was applied to identify enriched regions. H3K36me3 ChIP-Seq peaks were downloaded from a previous study (12). All of the ChIP peaks were quantified and viewed using the HOMER toolkit (43), cluster 3.0 (Michael Eisen/Michiel de Hoon) and Java Tree View (Alok J. Saldanha).

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